

Error analysis of the calibration of a dual-rotating-retarder Mueller matrix polarimeter

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We previously developed a Mueller matrix microscope by combining a dual-rotating-retarder device with a commercial microscope, which holds promise for pathologic diagnosis applications. In practical applications, the signals to be measured are sensitive to systematic errors, and the measurement accuracy and the instrument calibration affect the capacity to characterize the data. In conventional calibration, the initial positions of the rotating devices are constrained, causing calibration instability. To improve the calibration, we examine the mathematical model of the calibration process using previous studies, and derive a calibration algorithm with reduced error magnification. Our experimental results show the proposed algorithm improves calibration accuracy, and provides more robust results. © 2017 Optical Society of America

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1. INTRODUCTION

The Mueller matrix of a sample can provide abundant information concerning its microstructure, and recent studies have indicated that some information regarding the microstructural changes in organs during carcinogenesis can be characterized by the Mueller matrix [1,2]. Our research group has developed a Mueller matrix microscope by combining a dual-rotating-retarder device with a commercial microscope [3], and the experimental results on human cervical and liver cancerous tissues show that this instrument has good potential for application in pathologic diagnosis.

Some transformed polarization parameters, including Mueller matrix polar decomposition parameters and Mueller matrix transformation parameters [4,5], have been developed to characterize cancer-related information [6]. These parameters are derived from proper combinations of the Mueller matrix elements, and small misalignments or imperfections of the polarizing elements can introduce errors in measurement of the resulting Mueller matrices. Therefore, the parameters of interest are highly sensitive to systematic errors, and accurate calibration is especially important for this instrument.

The imaging section layout in a Mueller matrix microscope is designed not to be influenced by the polarization characteristics, so that its calibration can be regarded independently as that of a normal microscope; while the calibration of the polarization modulation section layout is the main challenge for a Mueller matrix microscope. The dual-rotating-retarder polarimeter, as first introduced by Azzam [7], is a basic Mueller

matrix measuring method that is widely accepted, owing to its simple configuration. Many scholars have studied the optimization of measurement schemes and the calibration of systematic errors. The optimization of the rotation ratio and the control of retarders were investigated by Smith [8], and additional discussions concerning factors such as the numbers of samples in a full measurement can be found in [9]. One common system configuration applies a measurement scheme in which the rotation ratio is set at 1:5, the sampling number is set as 30, and quarter-wave plates are applied as retarders. This system configuration is adopted in our instrument.

To calibrate systematic errors in the dual-rotating-retarder configuration described above, Hauge developed a method that can calibrate diattenuation errors in wave plates [10]. Chipman *et al.* introduced a system error model containing misalignments of the polarizing elements and imperfections in retardation of the retarders [11,12]. These researchers presented a calibration method based on the Fourier transform. The maximum errors for the absolute values of the Mueller matrix elements can be reduced to 0.01 when adopting this calibration method in our instrument. However, the calibration results fluctuate greatly when the initial fast axis directions of the retarders change. Therefore, two or more calibration operations are needed to achieve a precise calibration. Broch *et al.* investigated the systematic errors in detail at the first and second order, and demonstrated a calibration method based on four-zone averaging measurement [13,14]. Some types of errors that are related to the initial positions of polarizing elements are

overcome by this method, but other errors remain. To improve the robustness of this calibration method, we further examine the mathematical model of the calibration process on the basis of previous studies, and we derive a calibration algorithm with reduced errors in magnification. Experimental results show that the proposed calibration algorithm can increase the accuracy of calibration and improve robustness.

2. MATERIALS AND METHODS

A. Experimental Setup

As shown in Fig. 1, a fundamental dual-rotating-retarder Mueller matrix polarimeter is set up in our laboratory. This polarimeter consists of a light source, a polarization state generator (PSG), a polarization state analyzer (PSA), and a light intensity detector. The light source used in our configuration is a customized collimated LED light source with a center wavelength of 633 nm, and the bandwidth of the light source is about 10 nm. The PSG is composed of a fixed linear polarizer (GCL-050003, Daheng Optics, China) and a highly precise rotatable quarter-wave plate (WPQ10ME-633, Thorlabs, USA). The PSA consists of the same polarizing elements, but arranged in inverse order. A 14-bit CCD camera (Retiga R6, QImaging, USA) is used as the light intensity detector, and the readout noise is about seven electrons. The incident light emitted by the light source successively passes through the PSG, the sample, and the PSA, before being detected by the CCD. Two quarter-wave plates are installed in the electrical rotating platform (KPRM1E/M, Thorlabs, USA), which has an angle control precision of 1 arc sec. We rotate the wave plates in discrete steps to modulate the polarization state of the light. In the process of measurement, the wave plate in the PSG rotates 6° per step, and the wave plate in the PSA rotates 30° per step. Thirty rotations and intensity detections are executed to complete a full measurement, and the Mueller matrix elements can be computed from the detected intensities. The system shown in Fig. 1 is a fundamental configuration for the dual-rotating-retarder Mueller matrix polarimeter, to discuss the calibration performance. Thus, the following experimental results are executed from the intensities averaged on the whole detector with no imaging section in the system.

B. Mathematical Model

The process of Mueller matrix measurement by a dual-rotating-retarder polarimeter has been well summarized as a process of

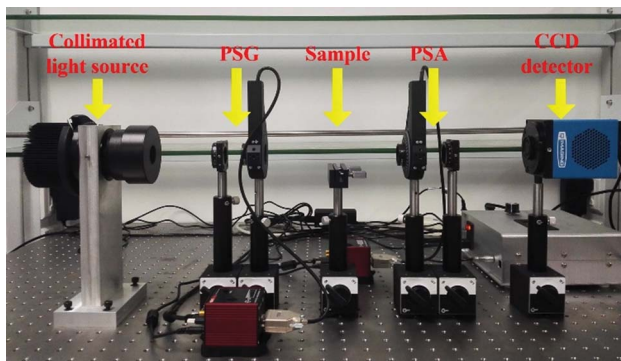


Fig. 1. Dual-rotating-retarder polarimeter configuration.

establishing and solving the linear equations described in Eq. (1) [15]. In this equation, M is a 16×1 vector that consists of all Mueller matrix elements, P is an $N \times 1$ vector that consists of the light intensities detected, W is an $N \times 16$ polarimetric measurement matrix, and N is the sampling number, which is 30 in our measurement scheme:

$$WM = P. \quad (1)$$

Misalignments and imperfections of the polarizing elements are the main sources of systematic error for the dual-rotating-retarder polarimeter, and five systematic parameters, namely α_2 , α_3 , α_4 , δ_2 , and δ_3 are used to characterize these systematic errors [11]. The definitions of these systematic parameters are shown in Fig. 2, where α_2 is the angle between the fast axis of the first quarter-wave plate and the principal axis of the first polarizer at the initial time, α_3 is the angle between the fast axis of the second quarter-wave plate and the principal axis of the first polarizer at the initial time, α_4 is the angle between the principal axes of the first and second polarizers, and δ_2 and δ_3 are the measures of actual retardation of the first and the second wave plates. The default values for α_2 , α_3 , α_4 , δ_2 , and δ_3 are 0° , 0° , 0° , 90° , and 90° . These five systematic parameters should be set close to the default values and known exactly in advance so that the Mueller matrix can be measured.

In the calibration process, a series of rotations and light intensity detections are executed with some standard samples, such as air, and the five systematic parameters are computed from the detected intensities. Denote $\vec{u} = (\alpha_2, \alpha_3, \alpha_4, \delta_2, \delta_3)^T$ as the vector of the five systematic parameters, and the calibration can be conducted as a process of establishing and solving the nonlinear equations described in Eq. (2), in which F describes the nonlinear mapping relationship between $\vec{u}$ and P :

$$F(\vec{u}) = P. \quad (2)$$

C. Calibration Algorithm

Typically, calibration equations are derived from the discrete Fourier coefficients of P . As the derivation of the algorithm has been explained clearly in [11], we quote their derivations directly in the paper and give a brief explanation. In theory, there are 11 nonzero Fourier coefficients: a_0 , a_2 , a_4 , a_6 , a_8 , a_{10} , b_2 , b_4 , b_6 , b_8 , and b_{10} , when air is used as the standard sample. The relationship between the error parameters and the 11 nonzero Fourier coefficients is detailed in Table 1. In that case, the systematic parameters can be calculated by Eq. (3):

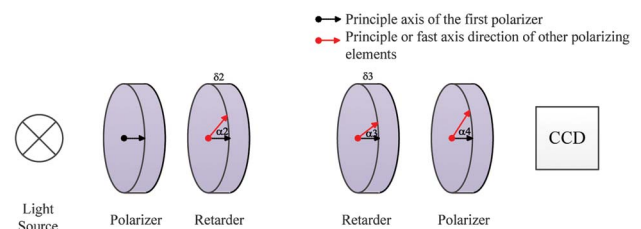


Fig. 2. Definitions of the systematic parameters.

Table 1. Relationship Between the Systematic Parameters and the Nonzero Fourier Coefficients

Nonzero Fourier Coefficients	
a_0	$\frac{1}{4} + \frac{1}{16} \cos(2\alpha_4)[1 + \cos(\delta_2)][1 + \cos(\delta_3)]$
a_2	$\frac{1}{16} \cos(4\alpha_2 - 2\alpha_4)[1 - \cos(\delta_2)][1 + \cos(\delta_3)]$
a_4	$-\frac{1}{8} \cos(2\alpha_2 - 2\alpha_3 + 2\alpha_4) \sin(\delta_2) \sin(\delta_3)$
a_6	$\frac{1}{8} \cos(2\alpha_2 + 2\alpha_3 - 2\alpha_4) \sin(\delta_2) \sin(\delta_3)$
a_8	$\frac{1}{16} \cos(4\alpha_2 - 4\alpha_3 + 2\alpha_4)[1 - \cos(\delta_2)][1 - \cos(\delta_3)]$
a_{10}	$\frac{1}{16} \cos(4\alpha_3 - 2\alpha_4)[1 + \cos(\delta_2)][1 - \cos(\delta_3)]$
b_2	$-\frac{1}{16} \sin(4\alpha_2 - 2\alpha_4)[1 - \cos(\delta_2)][1 + \cos(\delta_3)]$
b_4	$\frac{1}{8} \sin(2\alpha_2 - 2\alpha_3 + 2\alpha_4) \sin(\delta_2) \sin(\delta_3)$
b_6	$-\frac{1}{8} \sin(2\alpha_2 + 2\alpha_3 - 2\alpha_4) \sin(\delta_2) \sin(\delta_3)$
b_8	$-\frac{1}{16} \sin(4\alpha_2 - 4\alpha_3 + 2\alpha_4)[1 - \cos(\delta_2)][1 - \cos(\delta_3)]$
b_{10}	$-\frac{1}{16} \sin(4\alpha_3 - 2\alpha_4)[1 + \cos(\delta_2)][1 - \cos(\delta_3)]$

$$\begin{aligned}\alpha_2 &= \frac{1}{4} \arctan\left(\frac{b_8}{a_8}\right) - \frac{1}{4} \arctan\left(\frac{b_{10}}{a_{10}}\right), \\ \alpha_3 &= \frac{1}{2} \arctan\left(\frac{b_2}{a_2}\right) - \frac{1}{2} \arctan\left(\frac{b_6}{a_6}\right) + \frac{1}{4} \arctan\left(\frac{b_8}{a_8}\right) \\ &\quad - \frac{1}{4} \arctan\left(\frac{b_{10}}{a_{10}}\right), \\ \alpha_4 &= \frac{1}{2} \arctan\left(\frac{b_2}{a_2}\right) + \frac{1}{2} \arctan\left(\frac{b_8}{a_8}\right) - \frac{1}{2} \arctan\left(\frac{b_{10}}{a_{10}}\right), \\ \delta_2 &= \arccos\left[\frac{a_{10} \cos(4\alpha_3 - 4\alpha_2 - 2\alpha_4) - a_8 \cos(4\alpha_3 - 2\alpha_4)}{a_{10} \cos(4\alpha_3 - 4\alpha_2 - 2\alpha_4) + a_8 \cos(4\alpha_3 - 2\alpha_4)}\right], \\ \delta_3 &= \arccos\left[\frac{a_2 \cos(4\alpha_3 - 4\alpha_2 - 2\alpha_4) - a_8 \cos(4\alpha_2 - 2\alpha_4)}{a_2 \cos(4\alpha_3 - 4\alpha_2 - 2\alpha_4) + a_8 \cos(4\alpha_2 - 2\alpha_4)}\right].\end{aligned}\quad (3)$$

As described in Section 2.B, the calibration process is actually a process of establishing and solving nonlinear equations between $\vec{u}$ and P . To solve the overdetermined nonlinear equations, some overdetermined information is abandoned, and an analytic solution of the sub-equations is used as an approximate solution in the conventional calibration algorithm. With this algorithm, we can calibrate our instrument to a relatively high degree of accuracy. The calibration performance has been demonstrated in [3], and the maximum errors for the absolute values of the Mueller matrix elements can be reduced to 0.01. However, the data stability of this calibration algorithm has not been discussed, and it needs to be improved.

As shown in Eq. (3), only eight Fourier coefficients are used to calculate the five systematic error parameters. Therefore, some information is omitted when using this algorithm, and this information loss results in a loss of accuracy in the calibration results. A greater difficulty is that Eq. (3) includes many anti-trigonometric functions. As there are several anti-trigonometric functions for each map function, the expressions in the anti-trigonometric functions must be limited in the range of $(-90^\circ, 90^\circ)$ to ensure the uniqueness of the calibration results. This limitation results in certain constraints on the initial axis directions of the polarizing elements. The constraints on α_2 and α_3 when α_4 is equal to 0° and the retarders are perfect

quarter-wave plates are demonstrated in Fig. 3. As shown in the shadow area, α_2 and α_3 are limited in a small portion within $(-22.5^\circ, 22.5^\circ)$ for the validation of the conventional calibration algorithm. However, the fast axis directions of the two wave plates would change as the electrical rotating platform work in the dual-rotating-retarder device and, as a result of error accumulation or some other systematic reasons, the fast axis directions at the initial time often deviate from the default position in practical applications. In that case, the equations in Eq. (3) are invalid, and several attempts need to be performed to adjust the two variable parameters, so that they are controlled in the allowable range. In addition, the systematic parameters α_2 , α_3 , and α_4 are calculated first, and parameters δ_2 and δ_3 are calculated later on the basis of α_2 , α_3 , and α_4 . Therefore, there are two propagations of error in the calibrations for δ_2 and δ_3 . All of these factors together cause instability in the calibration results.

The robustness of the calibration algorithm can be improved if the optimal numerical solutions of the original equations are adopted as the calibration results. There is usually no exact solution for the original equations in Eq. (2). It is because there are modeling errors in the process of establishing Eq. (2), and the light intensity P measurement is not completely accurate. That is also the reason why an analytic solution of the sub-equations is used as an approximate solution in the conventional calibration algorithm. However, we can always find the least-squares solution $\vec{u}'$, which minimizes $\|F(\vec{u}) - P\|_2$, as the optimal numerical solution for the original equations. If we regard the expression $H(\vec{u}) = \|F(\vec{u}) - P\|_2$ as a function of $\vec{u}$, then $\vec{u}'$ is the minimum point of this function, so the derivative function of $H(\vec{u})$ at $\vec{u}'$ should be zero. Simplify the expression of the derivative function of $H(\vec{u})$ and then we get Eq. (4), in which $F'(\vec{u}')$ designates the Jacobian matrix of $F(\vec{u}')$:

$$[F'(\vec{u}')]^T \cdot [F(\vec{u}') - P] = 0. \quad (4)$$

Comparing with Eq. (3), Eq. (4) yields a first-order approximation of the original equations, so that less information is omitted. In addition, the nonlinear Eq. (4) can be solved by

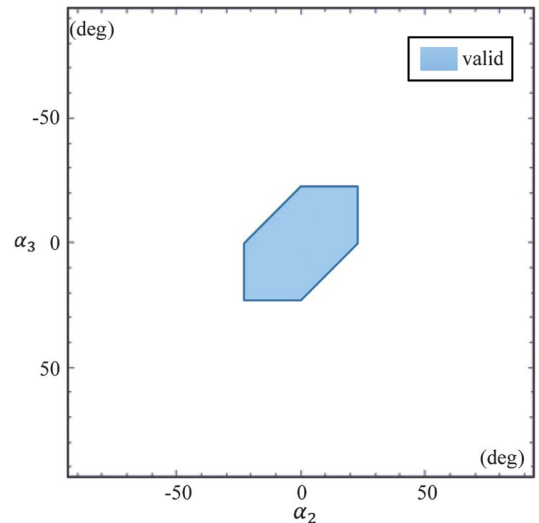


Fig. 3. Constraints on α_2 and α_3 when α_4 is equal to 0° and the retarders are perfect quarter-wave plates, in the conventional calibration algorithm.

iteration, and thus no anti-trigonometric function appears during the operations, so that no additional constraint is introduced. All systematic parameters are calculated simultaneously with the iteration method, and so there are no additional propagations of errors.

D. Error Propagation

The error propagation coefficients are used to describe the error propagations from the detected intensities to the calibrated systematic parameters, and can help to evaluate the performance of the calibration algorithms more intuitively.

In the conventional calibration algorithm, the systematic parameters are expressed in the form of Fourier coefficients, and the Fourier coefficients can be expressed directly by the measured intensity values, using the expressions described in Eq. (5). Therefore, Eq. (3) can be rewritten by replacing the Fourier coefficients with the corresponding intensity expressions:

$$\begin{aligned} a_i &= \begin{cases} \sum_{j=1}^{30} \frac{P(j) \cos[2\pi i(j-1)/30]}{30}, & (i = 0) \\ \sum_{j=1}^{30} \frac{P(j) \cos[2\pi i(j-1)/30]}{15}, & (i \neq 0) \end{cases}, \\ b_i &= \sum_{j=1}^{30} \frac{P(j) \sin[2\pi i(j-1)/30]}{15}. \end{aligned} \quad (5)$$

This substitution is a bit lengthy, but it is straightforward. For the sake of convenience, we use a single letter G to describe the nonlinear mapping relationship between P and $\vec{u}$ in the substituted equations. Then, the substituted equations can be simply expressed as $G(P) = \vec{u}$, and the error propagation coefficients can be achieved by performing simple partial derivative operations on $G(P)$:

$$\Delta \vec{u}(i) = G'(P) \Delta P(j), \quad (6)$$

$$\begin{pmatrix} \Delta \alpha_2 \\ \Delta \alpha_3 \\ \Delta \alpha_4 \\ \Delta \delta_2 \\ \Delta \delta_3 \end{pmatrix} = \begin{pmatrix} \frac{\partial \alpha_2}{\partial P(1)} & \cdots & \frac{\partial \alpha_2}{\partial P(30)} \\ \vdots & \ddots & \vdots \\ \frac{\partial \delta_3}{\partial P(1)} & \cdots & \frac{\partial \delta_3}{\partial P(30)} \end{pmatrix} \begin{pmatrix} \Delta P(1) \\ \vdots \\ \Delta P(30) \end{pmatrix}. \quad (7)$$

Equation (6) shows the partial derivation result, which is rewritten in matrix form in Eq. (7), where $G'(P)$ is the Jacobian matrix of $G(P)$. The elements in $G'(P)$ are exactly the error propagation coefficients in the conventional calibration algorithm, and the elements in row i and column j correspond to the error magnification from $P(j)$ to $\vec{u}(i)$. Thus, the 2-norm of the i th row in $G'(P)$ can be regarded as the comprehensive error magnification for $\vec{u}(i)$.

For the proposed calibration algorithm, the optimal numerical solution $\vec{u}'$ is calculated by the iteration method, and it satisfies Eq. (4). Substitute $\vec{u}'$ and P with $\vec{u}_{\text{actual}} + \Delta \vec{u}'$ and $P_{\text{actual}} + \Delta P$ in Eq. (4), respectively, where $\vec{u}_{\text{actual}}$ is the true value of the systematic parameters, and P_{actual} is the true value of the light intensities on the detector. Then conduct a Taylor expansion, and we obtain Eq. (8):

$$\begin{aligned} & [F'(\vec{u}_{\text{actual}}) + o(\vec{u}_{\text{actual}})]^T \\ & \cdot [F(\vec{u}_{\text{actual}}) + F'(\vec{u}_{\text{actual}}) \Delta \vec{u}' - P_{\text{actual}} - \Delta P] = 0. \end{aligned} \quad (8)$$

Simplify Eq. (8) by ignoring the high-order small quantities such as $[o(\vec{u}_{\text{actual}})]^T \Delta \vec{u}'$ and $[o(\vec{u}_{\text{actual}})]^T \Delta P$ to obtain Eq. (9):

$$\Delta \vec{u}' = \{[F'(\vec{u}_{\text{actual}})]^T \cdot F'(\vec{u}_{\text{actual}})\}^{-1} \cdot [F'(\vec{u}_{\text{actual}})]^T \cdot \Delta P. \quad (9)$$

Equation (9) is similar to Eq. (6), and it describes the error propagations in the proposed calibration algorithm. Both Eqs. (6) and (9) can be rewritten in the same form of $\Delta u = B \Delta P$, where B is the matrix which consists of all the error propagation coefficients in the calibration algorithm, and the 2-norm of each row in B describes the comprehensive error magnification for one error parameter.

The 2-norms, with different values of the systematic parameters, are calculated as illustrated in Fig. 4. The plots (a), (c), (e), (g), and (i) are calculated using the classical formulas, and plots (b), (d), (f), (h), and (j) are calculated using the proposed method. The default value for $\vec{u}$ is $(0^\circ, 0^\circ, 0^\circ, 90^\circ, 90^\circ)^T$, and we make one systematic parameter deviate from the default in each plot to observe the relationship between the comprehensive error magnifications and the values of error parameters. Concerning about the constraints on the error parameters in the conventional calibration algorithm, the range of deviation is selected as $(-30^\circ, 30^\circ)$. The parameters that deviate from the default are $\alpha_2, \alpha_3, \alpha_4, \delta_2$, and δ_3 respectively, from the first row on the top to the last row on the bottom. In addition, the drawings (a), (b), (c), and (d) when α_2 or α_3 deviates from the default are drawn in logarithmic coordinates to better display the data.

It is shown in Figs. 4(a), 4(c), 4(e), 4(g), and 4(i) that some types of error magnifications fluctuate by a large amount, as the systematic parameters deviate from the default in the conventional algorithm. When α_2 deviates from the default, the comprehensive error magnifications of δ_2 and δ_3 stay equal while varying greatly, but the comprehensive error magnifications of α_2, α_3 , and α_4 do not change. When α_3 or α_4 deviates, only the comprehensive error magnification of δ_3 changes. When δ_2 or δ_3 deviate, all of the comprehensive error magnifications fluctuate. The comprehensive error magnifications describe the ratio of the error in detected light intensities to that of the calibration results. Thus, the phenomena in Fig. 4 indicate that the detected errors of light intensities will be increased by a large amount when some types of systematic parameters deviate from the default. Another crucial point is that the error magnifications of δ_2 and δ_3 are more sensitive to deviations than those of α_2, α_3 , and α_4 . It shows that it is in agreement with the previous analysis in Section 2.C. In the conventional calibration algorithm, α_2, α_3 , and α_4 are calculated first, and parameters δ_2 and δ_3 are calculated later on the basis of α_2, α_3 , and α_4 . Therefore, there are two error propagations in the calibration process of δ_2 and δ_3 , and the calibration results of δ_2 and δ_3 are less robust. Figures 4(a) and 4(c) are drawn in logarithmic coordinates for better clarity, because when α_2 or α_3 deviates from the default, the variation range of some types of error magnifications is very large, from several to hundreds. The results also propose the necessity of accurate control of α_2 and α_3 in the conventional calibration algorithm.

The comprehensive error magnifications for the proposed algorithm are demonstrated in Figs. 4(b), 4(d), 4(f), 4(h), and 4(j). When α_2 or α_3 deviates, there is no change in error magnification. When α_4 deviates, the comprehensive error magnifications of $\alpha_3, \alpha_4, \delta_2$, and δ_3 change in small degrees. When δ_2 or δ_3 deviates, all of the comprehensive error magnifications

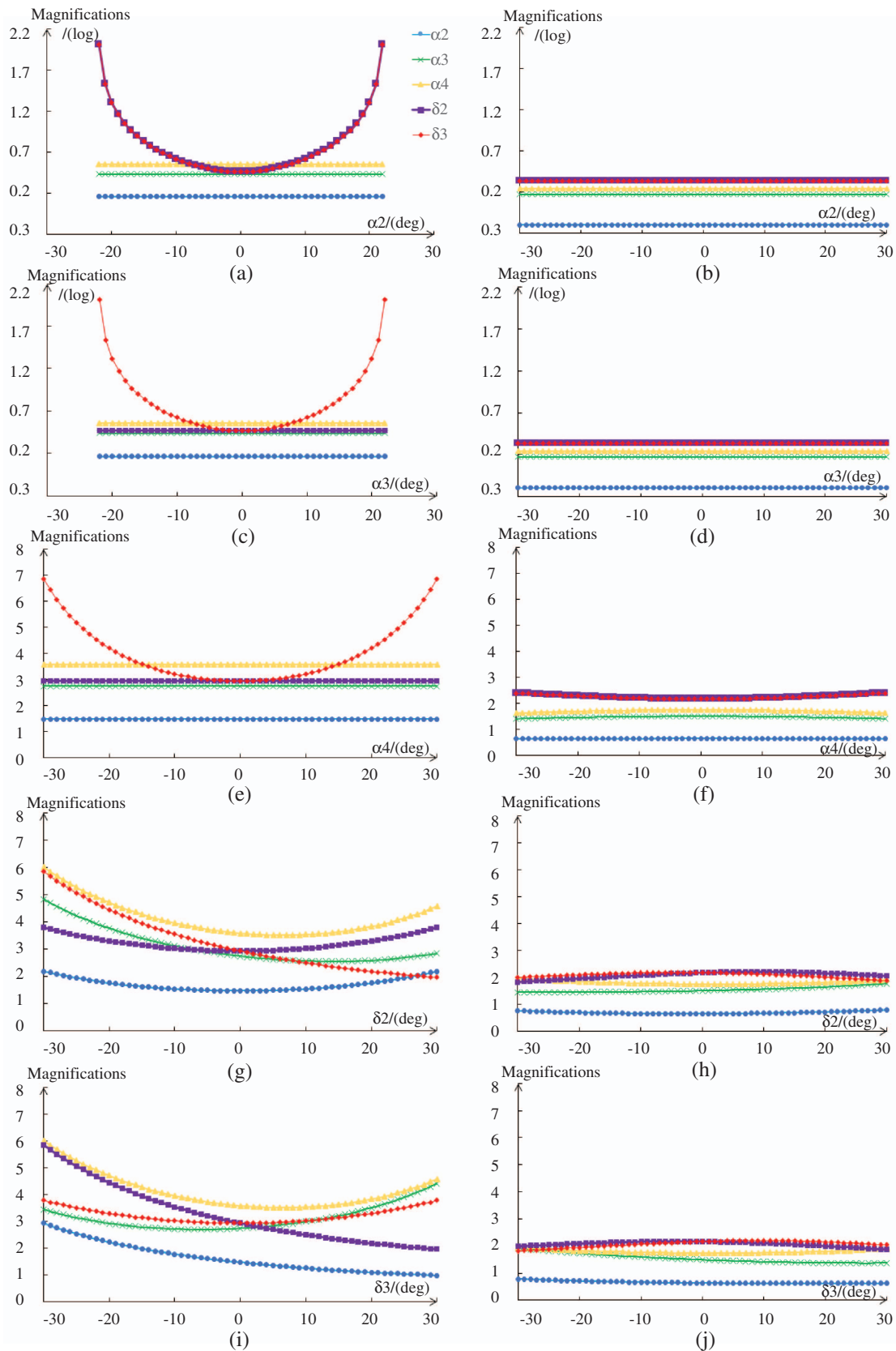


Fig. 4. (a),(c),(e),(g),(i): Comprehensive error magnifications in the conventional algorithm. (b),(d),(f),(h),(j): The proposed calibration algorithm where the systematic parameters deviate from the default.

fluctuate slightly. As we can see, both the absolute values and the degrees of fluctuation in the comprehensive error magnifications are smaller when the proposed algorithm is applied. These results can also offer support for the earlier analysis. It may be assumed that smaller error magnifications can be achieved when using a first-order approximation of all the original equations, while the conventional algorithm abandons some of the overdetermined information for simplifying expressions. The changes of error magnifications are small, even if the systematic parameters deviate as far as 30° from the default values. These results indicate that the robustness of the calibration algorithm does increase with the iteration method. Another point is that the variable systematic parameters α_2 and α_3 do not need to be limited in the range of $(-22.5^\circ, 22.5^\circ)$ and there is no error magnification change when α_2 and α_3 deviate. This outcome means that there are no additional constraints on the initial axis direction of the wave plates in the proposed calibration algorithm.

3. EXPERIMENTAL RESULTS

As α_2 and α_3 are the only two variable systematic parameters in the dual-rotating-retarder configuration, and as these variables are precisely controlled by the electrical rotating platform, we set α_2 and α_3 at several different values around the default values. Then we compare the performances of both the conventional and the proposed calibration algorithms.

The experimental process is as follows: (1) Execute the calibration process, and use the conventional calibration algorithm to obtain an appropriate calibration result. (2) Adjust the axial directions of the polarizing elements to compensate for errors in α_2 , α_3 , and α_4 . (3) Repeat steps (1) and (2) until α_2 , α_3 , and α_4

are very close to the default values. (4) Set α_2 to be -20° , -10° , 0° , 10° , and 20° , and execute the calibration process 10 times at each value. (5) Set α_3 to be -20° , -10° , 0° , 10° , and 20° , and execute the calibration process 10 times at each value. (6) Calculate the calibration results with both the conventional and the proposed calibration algorithms.

Figure 5 shows the average calibration results when α_2 or α_3 deviates from the default. Plots (a) and (c) are for the conventional algorithm, and plots (b) and (d) are for the proposed algorithm. The horizontal axis corresponds to the variable parameters, and it is α_2 in plots (a) and (b) and plots (c) and (d) corresponds to α_3 . The vertical axis shows the difference between the calibration values and the assumed values. The standard deviations of 10 calibrations with different values of α_2 and α_3 are also calculated, and are shown in Fig. 6.

These experimental results clearly show that the calibration results from the conventional algorithm are unstable when the variable parameters deviate from the default values, but the calibration results with the proposed algorithm have good consistency. The lines tendencies in Fig. 6 are in agreement with the theoretical results shown in Fig. 4. With the conventional algorithm, the errors from the detected light intensities are amplified greatly if α_2 or α_3 deviates from the default. When the proposed algorithm is applied, the error of light intensities is suppressed and only causes a small error less than 0.05° in the calibration results.

As shown in Figs. 5(a) and 5(c), the calibration results with different systematic parameters are deviated from each other, where the standard deviations of 10 calibrations with the same systematic parameter are relatively small. These phenomena indicate that there are certain error sources in the calibration

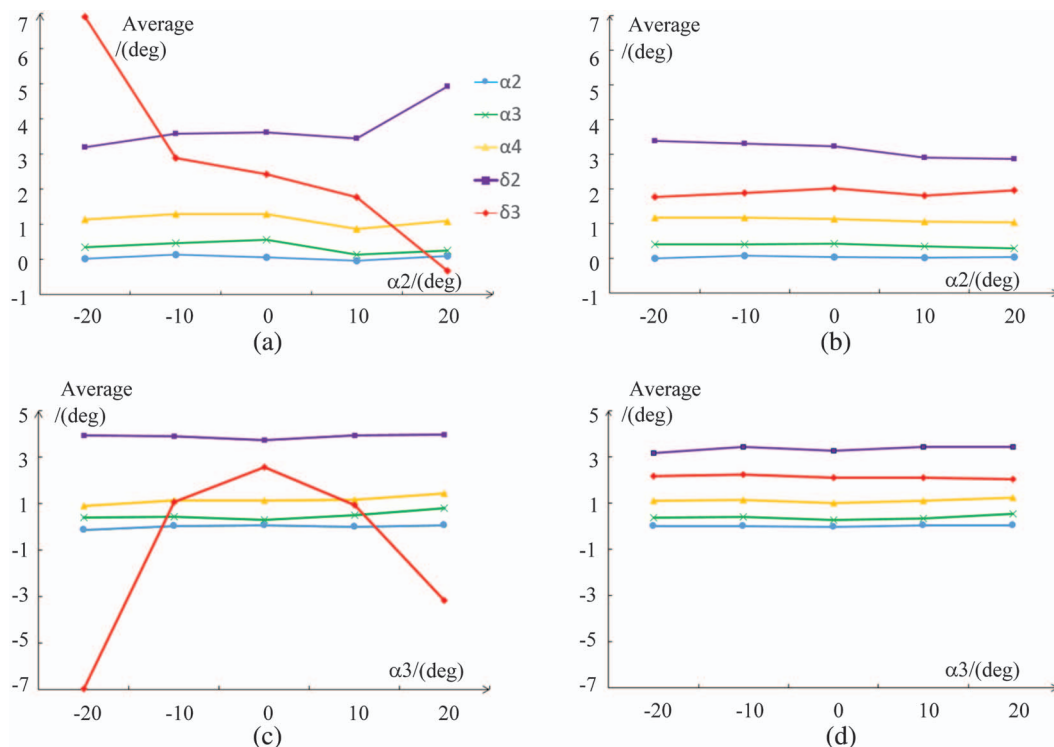


Fig. 5. Average calibration results for (a),(c) the conventional method, and (b),(d) the proposed method, when α_2 or α_3 deviates from the default.

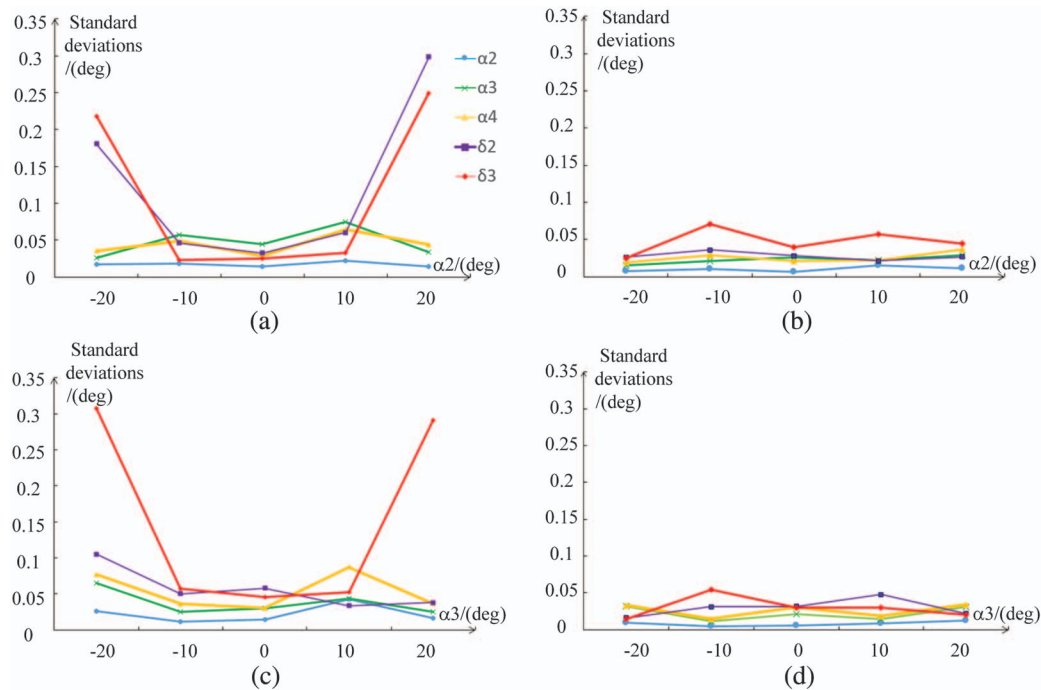


Fig. 6. Standard deviations of calibration results for (a),(c) the conventional method, and (b),(d) the proposed method, when α_2 or α_3 deviates from the default.

procedure other than error in the detected light intensities. They could be the certain modeling errors in the polarizing elements or something else that could introduce errors in the calibration results by using the conventional calibration algorithm. In the proposed algorithm, however, these unknown errors are also restrained.

4. CONCLUSION

We have reviewed the mathematical model of the calibration procedure for the dual-rotating-retarder Mueller matrix polarimeter, and proposed a calibration algorithm with reduced error magnification. The optimal numerical solution, instead of an analytic solution, is applied as the calibration result in the proposed calibration algorithm, and a fundamental dual-rotating-retarder polarimeter is set up to examine the performance of the proposed calibration algorithm. The comparison of the experimental results of the conventional and the proposed calibration algorithms indicated that fewer constraints on the variable parameters would be required in the proposed calibration algorithm. This improvement can help simplify the calibration operations in practical applications. As the discussions in this paper are partly an investigation of the Mueller matrix microscope instrumentation, the results in this paper can be directly used in the calibration of Mueller matrix microscopes. Our future work will concentrate more on the influences of imaging devices and the improvement of measurement accuracy in Mueller matrix microscopes.

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